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Project Title:	Use of Computed Tomography for Characterizing Materials Grown Terrestrially and in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	35812	**Congressional District:**	5
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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CT scanning is based on the ability to process millions of attenuation values resulting from the interaction of a high energy photon beam with a sample. By changing the direction of the incident beam, the algorithms can furnish the values for linear absorption coefficients in two dimensional space within the slice being irradiated. CT arrangements, including the one at KSC, typically use banks of many detectors, and are programmed to translate the sample followed by angular rotation and repetitive translation. By this means much statistical data are obtained. The technique has several important implications for use by the OBPR PIs. First, the system is capable of examining several square feet of sample within the narrow height selected. This means that many samples, and of very different types, can be examined simultaneously. Secondly, as the absorption values can be obtained in two dimensions, it becomes possible to obtain values while the samples are still encased. During the ATD project, satisfactory results were obtained for metallic samples while they were still within their protective Sample Ampoule Cartridge Assemblies (SACAs). Thus the

Task Description:	technique is capable of measuring density within not only a fused silica ampoule, but also while within the encasing inconel cartridge. With the exception of edge effects, the work is quantitative. The CT system at KSC is being used with a cobalt-60 source, as compared with the normal x-ray sources employed. Penetration is up to the equivalent of 5 inches of steel. To our knowledge this feature is unique world wide, and has the advantage of producing a monochromatic beam with no preferential absorption as happens with the x-ray generated beams. So-called beam hardening effects are thus eliminated. The CT system at Marshall Space Flight Center has also been used on occasion. The purpose behind this work is to provide NASA Flight and Ground Principal Investigators (PI) rapid information, non-destructively, about their samples. This information will be in the form of density values throughout the samples, especially within slices 1 mm high. With correct interpretation and good calibration, these values will enable the PI to obtain macro chemical compositional analysis for his/her samples. Alternatively, the technique will provide information about the porosity level and its distribution within the sample. Experience gained with a NASA MRD-sponsored Advanced Technology Development (ATD) project on this topic has brought the technique to a level of maturity at which it has become a viable characterization tool for many of the Materials Science PIs, but with equipment that could never be supported within their own facilities. The existing computed tomography (CT) facility at NASA's Kennedy Space Center (KSC) is ideally situated to furnish information rapidly and conveniently to Flight PIs, particularly immediately before and after flight missions. In addition, the measurement technique developed can be used for determining density values of samples within operating furnaces, and hence is useful for determining density and dilation in materials at elevated temperatures. The proposed work here is designed to be a service to current and future PIs, and as such will "will lead to the definition or enhance the understanding of existing or potential flight experiment in material science." It will provide PIs with a mechanism for rapidly ensuring the integrity of their SACAs on landing, and for determining macro-compositional values of samples. This can be done within 24 hours, and is non-destructive. Thus the technique will enable the PI to determine how best to open the SACA, and where to cut the sample. It can also be used for providing density measurements for elements and compounds through temperatures from room temperature to 900oC, through the use of a custom-designed furnace which can be used in situ on the CT translating table. Such data are proving invaluable for PIs who are calculating convection in their samples and require accurate thermophysical constants, and by engineers who are designing SACAs and require thermal expansion information.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Work continued on two fronts, namely the examination of meteorite samples and the interpretation of the data, and the dilation of metallic and semiconductor samples within the Marshall Space Flight Center (MSFC)-designed and built furnace. Data on the Mundrabilla were analyzed in conjunction with Professor Voorhees' team at Northwestern University, in particular with Roberto Mendoza, who obtained his Ph.D. in 2004. The work uses quantitative stereology to describe three-dimensional structures. When considering Mundrabilla as a two-phase structure of Fe-Ni and FeS, the data gave a volume percent of 72% for the FeS (troilite) phase. Stereology parameters were given giving an inverse surface area per unit volume of 8.141 mm. This compares with Mendoza's own work on Pb-Sn alloys where the longest coarsened result was about 100 microns. Principal curvature diagrams demonstrate that the predominant radius of curvature is 12.5 mm, and that the position of the peak within the field shows that saddle shapes are the main feature. Other data indicated that the FeNi phase is essentially interconnected and spans the entire length and width of the 50 kg sample. The analysis found only four separated FeNi regions, and the assumption can be made that with a larger sample, these also would have been connected. This implies that there was a unidirectional solidification with the troilite trapped within the solidifying front. Whether the mechanism was dendritic cannot be uniquely confirmed due to the coarsening that has taken place. The second meteorite examined was that discovered in Colomera, Spain. It is a IIE iron meteorite with alkali-rich silicate inclusions. As with Mundrabilla, these are of at least a cm size and readily imaged with CT. Previous characterization of this meteorite has been destructive with spherical glassy inclusions being literally carved out of the matrix. The CT scanning was performed with both the cobalt source at Kennedy Space Center (KSC) (Mr. Engel) and by Dr. Ron Beshears at MSFC with a high resolution x-ray system. The KSC analysis was done to provide a signature for a typical silicate density material to enable the finding of such minerals in Mundrabilla. Distribution of silicates with the two samples of Colomera were also determined, but the resolution was inferior to the data provided at MSFC. Several morphologies of silicates were observed, mainly spherical but with some elongated. The two samples were approximately 1 inch square by 3 inches long, and were small enough that beam hardening was not an issue. The results in the form of video sections have been provided to Professor Wasserburg at California Institute of Technology. Two more samples were heated, melted, and frozen in the heat pipe furnace mounted on the CT system. Digital radiographs (DR) were taken at all temperatures, and CT scans were made at a minimum of three locations on each. Digital radiographs, which provide length data and hence thermal expansion, require careful evaluation as initially the load does not fill the ampoule and slumps during melting. The data in the liquid state are exceptional due to the low thermal expansion of the containment material – fused silica. Post solidification cooling DRs were useful to identify locations where pipes were formed. Re-heating was used to eliminate this. The CT scans clearly showed the difference in density, except during the mush stages, where top-to-bottom segregation was noticeable. The data in the solid and liquid states present clearly the density changes with respect to temperature. Extrapolation of the lines gives the dilation on melting. The possibility of fabricating a eutectic composition for further work is being considered. This would not suffer from any segregation difficulties. The results are in a final draft form prior to being submitted for publication.
Bibliography Type:	Description: (Last Updated: 09/19/2019)
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